

Jonathan Alexander Sanchez

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Education

University of Memphis

Expected December 2028

Bachelor of Science with Honors in Computer Science

Memphis, TN

- Concentration: Artificial Intelligence | Minor: Entrepreneurship
- GPA: 3.96

Technical Skills

Languages: Python (Proficient), C# (Proficient), HTML5/CSS3 (Proficient), JavaScript (Novice), C++ (Novice), MATLAB (Novice)

Developer Tools: Git & GitHub, Pulsar & Visual Studio Code, Unity 6 (C#, URP), Blender

Certifications: CompTIA (ITF+)

Frameworks/Services: Flask, SMTP, Cesium for Unity, OpenXR / XR Interaction Toolkit, GCP API

Geospatial & Performance: WGS84/ECEF Coordinate Systems, Performance Optimization

Miscellaneous: Prompt Engineering, Arduinos & General Electronics

Experience

VR Fault & Seismicity Visualization Developer

September 2026 – Present

Center for Earthquake Research and Information, University of Memphis

Memphis, TN

- Shipped six standalone Windows builds (three desktop, three OpenXR-based PC VR) across three earthquake-catalog detail tiers, including reduced M2+/M3+ datasets that cut rendered objects up to 98.5% for low-end hardware
- Extended the application to a second fault system, the New Madrid Seismic Zone spanning six states, adding depth-colored fault surfaces, an 11,000+ event catalog, and region-aware movement and population controls
- Built a seismic-station layer with procedural 3D markers for 100+ stations that load and unload within a 50 km radius, and added 400+ city labels
- Refined earthquake-cube sizing in the geometry pipeline for more accurate magnitude representation
- Fixed build and reliability bugs, including frame-rate-dependent input loss, shader stripping, coordinate drift, and ECEF-related terrain clipping; added a two-step VR quit guard

VR Fault & Seismicity Visualization Intern Developer

June 2026 – August 2026

Center for Earthquake Research and Information, University of Memphis

Memphis, TN

- Built a Unity 6 application from scratch that streams photorealistic 3D terrain (Cesium for Unity) and overlays real fault and seismicity datasets with Desktop and VR compatibility
- Designed and implemented a geospatial data pipeline (MATLAB, Blender, Unity) that converts raw fault and seismic data (400+ faults and 1.6M+ earthquakes) into 3D mesh geometry
- Created an artificial underground mode that was not supported by Cesium at the time of development
- Developed a region-switching system, a runtime settings UI (dropdowns, toggles, data layers), and a city label system optimized for performance
- Iterated features through testing cycles, keeping detailed documentation of design decisions, bugs, and fixes from the project's development

Personal Projects

Automated Invoice Generator

May 2025 – August 2025

Python, Google Sheets API, SMTP, FPDF

- Created a fully automated system that connects Google Forms, Google Sheets, and Gmail to create and send invoices to clients
- Collected work details, descriptions, and costs through a Google Form synced in real time to a Google Sheet
- Automated invoice delivery, removing the need for manual file handling

Partner Management System

January 2024 – June 2024

Python, Flask, Google Cloud Platform API, HTML/CSS

- Developed and designed a full-stack web application to manage organizational partner data as a replacement for manual spreadsheet management
- Stored partner data persistently in a local JSON file with full CRUD support
- Integrated Google Sheets API to automatically update partner data on every change for real-time backup
- Implemented quality-of-life search and filter features into the user interface to improve user experience
- Created a custom audit system that records all user actions with timestamps as a form of data integrity